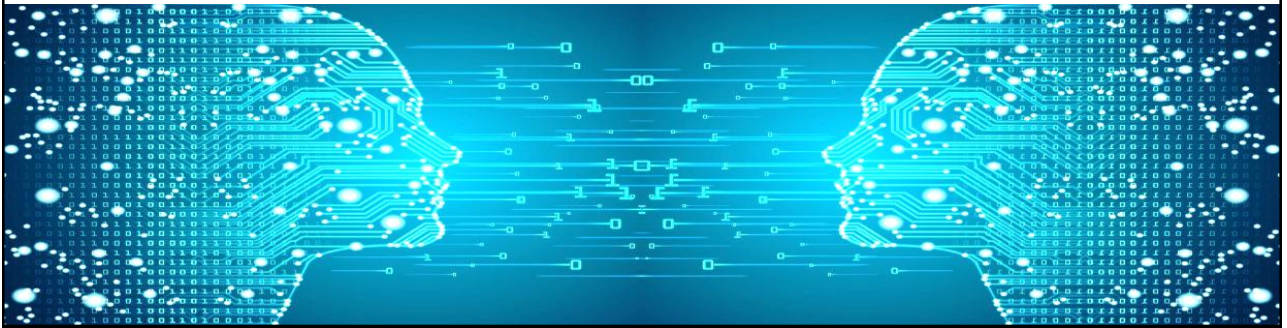
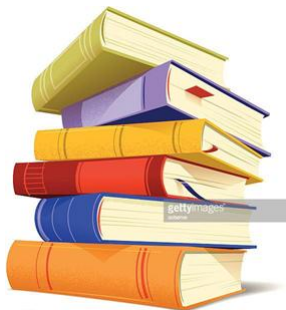


Machine Learning



Hidden Markov Models (HMM)



- Hidden Markov Model (HMM)
- The Three Basic Problems of HMM
- HMM Applications
- Example

Hidden Markov Model (HMM)

- A Hidden Markov Model is a probabilistic model consisting of:
 - **Hidden states:** not directly observed
 - **Observations:** emitted by hidden states
 - **State transitions:** Markov chain
 - **Emission probabilities:** distribution over observations

The Three Basic Problems of HMM

- The primary tasks solved using HMM fall into three basic categories:
- 1. Scoring Problem (Evaluation):**
 - **Goal:** To find the probability of a specific observed sequence given that the HMM is already defined.
 - 2. Alignment Problem (Decoding/Path Finding):**
 - **Goal:** Given an observation sequence and a defined HMM, determine the **most likely sequence of underlying hidden states** that might have generated it.
 - 3. Training Problem (Learning):**
 - **Goal:** To create or adjust the parameters (probabilities) of an HMM given a particular set of related training sequences.

HMM Applications

- Speech Recognition.
- Handwriting and Gesture Recognition.
- Part-of-Speech Tagging.
- Bioinformatics.

Scenario: Guessing the Boss's Attitude

- An employee's behavior at home is strongly influenced by his boss's attitude at work.
- His wife wants to infer what happened at the office based on his reactions at home, but she has very limited information about the workplace.
- Assumptions:
 - The boss has one of two attitudes each day: angry or happy.
 - The employee shows one of three reactions at home: keep silent, constantly yell, or talk happily.
- The wife, who is skilled in ML, decides to use an HMM to guess the boss's attitude over a sequence of days.
- She constructs an HMM representation with hidden states (boss's attitudes) and observations (employee's reactions) to make a scientific inference.

Scenario: Guessing the Boss's Attitude

▪ A. Hidden States (boss's attitudes)

- State 1: **Angry**
- State 2: **Happy**

▪ B. Observations (Employee's Reactions)

- Observation 1: **Silent**
- Observation 2: **Yelling**
- Observation 3: **Joking**
- Observation Sequence (four consecutive days): Silent, Yelling, Joking, Silent.

Parameters

▪ Start probability (π):

The Boss's Attitude	Probability
Angry	0.4
Happy	0.6

▪ Transition probability (A):

The boss's attitude change	Probability
Angry → Angry	0.6
Angry → Happy	0.4
Happy → Angry	0.3
Happy → Happy	0.7

Parameters

■ Emission probability (B):

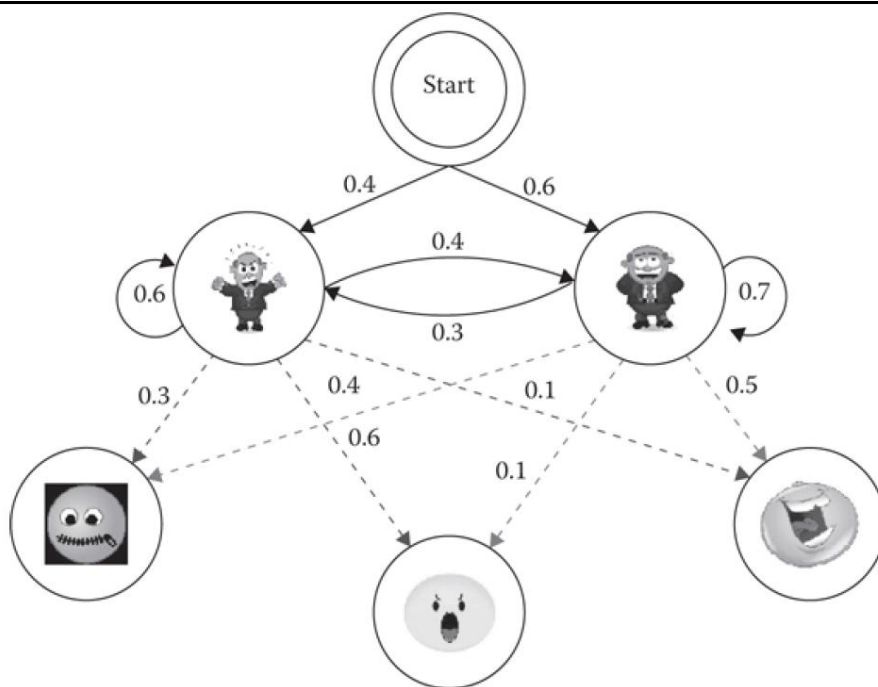
- Emission probabilities representing relationship between states and observation.

Boss State: Angry

Employee State:
Silent (0.3), Yelling (0.6), Joking (0.1)

Boss State: Happy

Employee State:
Silent (0.4), Yelling (0.1), Joking (0.5)



Alignment Problem: Path Calculation Logic

▪ Goal: Find the Most Likely Hidden State Sequence

- The task is to find the sequence of hidden states Q that maximizes the probability of the path given the observation sequence O .
- This is achieved using a dynamic approach (Viterbi-like approach), calculated day by day:
 - Day 1 calculates the best probability of being in each state.
 - Day 2 uses the results of Day 1 to calculate the best probability.
 - ...
 - And so on, until the last day, we find the optimal path.

Alignment Problem: Path Calculation Logic

I. Day 1: Initialization (Observation: Silent)

- $V_1(\text{Angry}) = \pi_{\text{Angry}} \times B_{\text{Angry}}(\text{Silent}) = 0.4 \times 0.3 = 0.12$
- $V_1(\text{Happy}) = \pi_{\text{Happy}} \times B_{\text{Happy}}(\text{Silent}) = 0.6 \times 0.4 = 0.24$
- $V_1(\text{Angry})$ is the maximum probability of any path ending in Angry state at time $t = 1$, after observing Silent

Alignment Problem: Path Calculation Logic

II. Day 2 onwards: Recursion

$V_t(j)$ is the probability of the path with the highest probability that leads to state j at time t and produces a sequence of observations O_1 to O_t .

$$V_t(j) = \max_i [V_{t-1}(i) \times A_{i \rightarrow j}] \times B_j(O_t)$$

Alignment Problem: Path Calculation Logic

III. Final Result (After 4 days)

- After four days of calculations, the algorithm retains the maximum path probability. Two paths remain competitive:
 - Path 1: **Angry, angry, happy, happy, angry** (Probability: 0.000726).
 - Path 2: Angry, angry, happy, happy, happy (Probability: 0.000169).
- The most likely hidden state sequence is **angry, angry, happy, happy, angry**.

HMM Key Concepts

- HMM is a sequential model where the crucial dynamics (states) are **hidden**.
- The three types of core problems allow for **scoring** (likelihood), **decoding** (finding the optimal path/alignment), or **training** (learning parameters).
- The parameters required are: Start Probability (π), Transition Probability (A), and Emission Probability (B).
- The alignment problem is solved using a dynamic approach that retains the maximum probability path at each step, as demonstrated by the final result: angry, angry, happy, happy, angry.

HMM Key Concepts

- HMM remains a foundational tool in many fields:
 - **Speech Recognition:** Analyzing sequences of phonemes or acoustic features.
 - **Computer Vision:** Gesture recognition (modeling sequences of movements).
 - **NLP:** Part-of-Speech tagging.
 - **Genetics:** Bioinformatics (sequence analysis).

